Year 13 Further Maths - Hyperbolic functions

Similar questions - Identities and equations

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1

(a) Solve the equation $3\cosh x + 5\sinh x = 5$, giving your solution in exact logarithmic form. **[4]**

(b) In this part you must show detailed reasoning. Find $\int_{-1}^{1} (3\cosh x + 5\sinh x)\,\mathrm{d}x$ in the form $a\mathrm{e} + \frac{b}{\mathrm{e}}$ where a and b are integers. **[3]**

2 In this question you must show detailed reasoning.

(a) Using the definitions of $\cosh x$ and $\sinh x$ in terms of exponentials, show that $\cosh 2x \equiv 2\cosh^2 x - 1$. **[2]**

(b) Solve $\cosh 2x = \cosh x + 2$, giving all solutions in exact logarithmic form. **[5]**

(c) Solve $\sinh 2x = 3\sinh x$, giving all solutions in exact logarithmic form. **[4]**

Answers

Question 1

(a) $3\cdot\frac{e^x+e^{-x}}{2}+5\cdot\frac{e^x-e^{-x}}{2}=4e^x-e^{-x}=5$. With $t=e^x$: $4t^2-5t-1=0$, $t=\dfrac{5\pm\sqrt{41}}{8}$. Only the positive root: $x=\ln\dfrac{5+\sqrt{41}}{8}$

(b) $\int_{-1}^{1}(3\cosh x+5\sinh x)\,dx=[3\sinh x+5\cosh x]_{-1}^{1}=(3\sinh 1+5\cosh 1)-(-3\sinh 1+5\cosh 1)=6\sinh 1=3e-\dfrac{3}{e}$. So $a=3$, $b=-3$

Question 2

(a) $\cosh 2x=\frac{1}{2}(e^{2x}+e^{-2x})$ and $2\cosh^2 x-1=2\cdot\frac{1}{4}(e^{2x}+2+e^{-2x})-1=\frac{1}{2}(e^{2x}+e^{-2x})$

(b) $2c^2-1=c+2$ with $c=\cosh x$: $2c^2-c-3=0$, $(2c-3)(c+1)=0$. $c=-1$ is impossible ($\cosh x\geqslant 1$), so $\cosh x=\frac{3}{2}$ and $x=\pm\ln\left(\frac{3}{2}+\sqrt{\frac{9}{4}-1}\right)=\pm\ln\dfrac{3+\sqrt{5}}{2}$

(c) $2\sinh x\cosh x=3\sinh x$, so $\sinh x(2\cosh x-3)=0$. $\sinh x=0$ gives $x=0$; $\cosh x=\frac{3}{2}$ gives $x=\pm\ln\dfrac{3+\sqrt{5}}{2}$